

DecodeLabs Project 3:

SQL Data Analytics



PostgreSQL

Objective

The objective of this project was to use SQL to analyze a sales dataset and answer key business questions through data retrieval , filtering, sorting, grouping and aggregation. The project demonstrates how SQL can transform raw transactional data into business insights that support information decision-making.

SQL Workflow

S/N	Step	SQL Command
1	Data Retrieval	SELECT
2	Data Source	FROM
3	Data Filtering	WHERE
4	Data Grouping	GROUP BY
5	Data Aggregation	COUNT(), SUM(), AVG()
6	Data Sourcing	ORDER BY
7	Result Limiting	LIMIT

Business Questions Answered

1. How many total orders are in the dataset?
2. How many unique customers made purchases?
3. What is the total revenue generated?
4. What is the average order value?
5. Which products generated the highest and lowest sales?
6. Which payment method is used most frequently?
7. Which payment method generated the highest revenue?
8. Which referral source generated the most orders?
9. Which shipping address received the highest number of orders?
10. Which customers spent the most money?
11. Which products sold the highest quantities?
12. What are the top 10 highest-value orders?
13. How many orders belong to each order status?
14. Which orders used coupon codes?
15. What is the average quantity purchased per order?

Key Findings

Below are few of the output/ results discovered by my business questions queries.

- Total Orders: 1,200
- Unique Customers: 1,189
- Total Revenue: 1,264,761.96
- Average Order Value: 1,053.97
- Chair generated the highest sales, while Phone generated the lowest.
- Online was the most frequently used payment method.
- Credit Card generated the highest total revenue.
- Instagram was the leading referral source for customer acquisitions.

Conclusion

This project demonstrates the practical application of SQL in answering real-world business questions using structured data. The analysis revealed important insights into customer purchasing behavior, product performance, revenue generation, payment preferences, and marketing effectiveness. By applying SQL concepts such as SELECT, WHERE, GROUP BY, ORDER BY, COUNT(), SUM(), and AVG(), I successfully transformed raw sales data into actionable business information. This project strengthened both my SQL proficiency and my ability to interpret query results to support data-driven business decisions.